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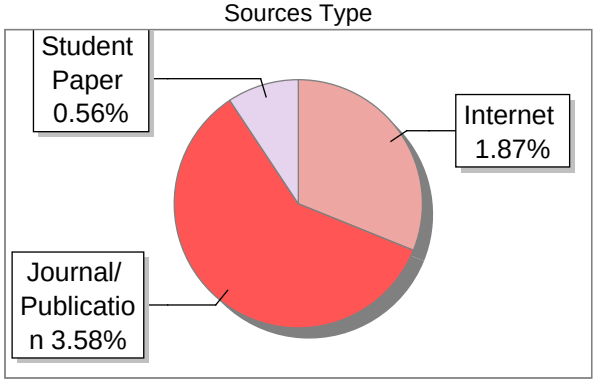

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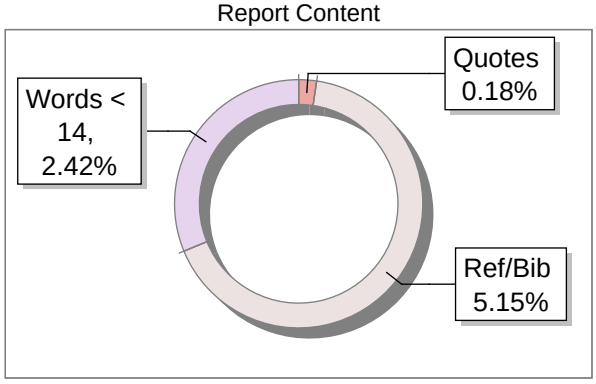

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Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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ABSTRACT

Retrieval Augmented Generation significantly improves Large Language Models but faces issues involving complicated syntactic ambiguities and loss of data when creating knowledge graphs. To tackle these problems, the Quantum Retrieval Augmented Generation is suggested in the context of Farm Fetching architecture by combining the efficient classical pipeline designed for linear processing with the quantum processor responsible for non-linear structure analysis. First, a classical approach evaluates 130,000 documents in terms of MongoDB vector search, dynamic Neo4j Knowledge Graphs, and baseline language model. This part of the research involves the introduction of Constructive Hallucination, a technique designed to validate and include artificial objects in order to enhance dynamic knowledge graphs. While various tests, including Thematic Correlator, showed superior performance of classical models in terms of semantics over the quantum systems, the classical parser cannot deal with deep structural ambiguity because of local attachment bias. For these difficult situations, a Quantum NLP Module performs an Ambiguity Controller, which is responsible for the redirection of requests. Using grammatical dependencies as input, quantum parameters are built into the quantum circuit, which uses entanglement to process the non-linear structures. The accuracy gain achieved in this way was 46 percent compared to the classical heuristic solution. Therefore, the Utility Threshold was identified, meaning that the combination of classical efficiency and quantum-nativeness provides a solid foundation for future studies' evaluation systems.

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CHAPTER 1

INTRODUCTION

1.1 Background

In the domain of NLP, large language models (LLMs) have made considerable progress. Large language models demonstrate a remarkable ability in text generation, summarization, and handling complex questions. However, these models depend greatly on parametric knowledge gained through training, thus limiting their ability to access recent information or specialized knowledge, causing problems such as hallucination and inaccuracy.

One possible solution to the above-mentioned problems is the use of the retrieval-augmented generation (RAG) paradigm. In RAG, LLMs are supplemented with external knowledge. Traditionally, documents are first retrieved from the vector database (e.g., MongoDB), and then used as the context for generation. Vector-based search can retrieve semantically similar documents; however, it may struggle to obtain contextual relations. On the other hand, dynamic Knowledge Graphs (KGs), using platforms like Neo4j, can directly incorporate the relational context between entities.

1.2 Problem Statement

In spite of all the advantages of retrieval-augmented generators, the efficiency of such a model is greatly influenced by the precision and depth of the retrieval mechanism. Analyzing a large data corpus, for instance, a knowledge base with 130,000 elements, two problems characteristic of current architectures become apparent.

The first one is related to the fact that classical graph extraction algorithms usually disregard any pieces of information that do not fit into the existing schema. As a result, the missing links that the large language model tries to add turn out to be ”hallucinations,” and the constructed Knowledge Graph appears to be sparse and fragmented.

Another problem is associated with the inability of the current architecture to analyze

syntactically ambiguous sentences. For instance, the current algorithms are based on simple keyword matches, vector similarity measures, and linear heuristic parsers. Classical natural language processing cannot overcome the tendency to commit errors under the condition of local attachment biases. It means that the decisions regarding parsing will be taken not considering the context but solely because of the proximity of syntactic elements. In particular, it poses a challenge in the presence of deep prepositional phrase attachments and reduced relative clauses.

1.3 Motivation

In this regard, the investigation mainly follows the Farm-Fetching principle, whereby it identifies the utility criteria that different computational models need to satisfy. Classical heuristics serve as a very efficient mechanism of standard linear searches for regular semantic information. However, integrating the mechanisms used in the classical model of computation through ”Constructive Hallucination” allows for the augmentation of knowledge graphs to facilitate their evolution into their most potent state.

However, the classical efficiency becomes limited in the face of an ”exotic farm” that consists of complex syntactic dependencies and nonlinear language systems. The incapability of the RAG model based on the classical framework to distinguish these complex syntactic nuances necessitates the development of a specific computational shift. In this regard, quantum computing emerges as a viable alternative owing to its ability to represent and process high dimensional data in the Hilbert space. With the principles of superposition and entanglement, quantum-based algorithms are capable of modeling the complex nonlinear interdependencies that occur in human language, leading to classical parsing failures.

1.4 Proposed Approach

QRAG represents an architecture that merges large-scale classical retrieval with quantum precision in two phases, as follows:

First, a strong classical foundation is laid. In this regard, a semantic search engine is employed for a retrieval task using more than 130,000 documents. Here, MongoDB Vector

Search, dynamic Neo4j Knowledge Graphs, and a baseline LLM are tested against each other. To address the problem of classical information loss, a Constructive Hallucination approach is proposed to detect and verify LLM hallucinated entities, thus dynamically enriching the Neo4j database.

Secondly, the quantum co-processor is introduced. The ambiguity controller is used as a routing algorithm. Queries that are linear can be easily answered using a strengthened classical pipeline. For structurally complex queries, however, they are routed to a Quantum NLP pipeline that uses variational quantum circuits running both on simulated and real IBM 127 qubits quantum hardware.

1.5 Contributions

The key contributions made through this study include the following:

- Construction and testing of a classical RAG architecture using 130,000 documents for comparing the performance of MongoDB vector search, dynamic Neo4j Knowledge Graphs, and benchmark large language models.
- Proposing and implementing the constructive hallucination approach to validate the creation of synthetic entities to enhance dynamic Knowledge Graphs.
- Establishment of syntactic ambiguity and attachment bias as the major limitations associated with classical RAGs.
- Formulation of the Farm-Fetching framework that defines the utility point where classical becomes quantum natural language processing problems.
- Creation of a combined Quantum and Classical computing platform called QRAG using an Ambiguity Controller to facilitate dynamic query routing.
- Implementation of Variational Quantum Classifier (VQC) techniques to encode grammatical dependency structures into parameterized quantum circuits.
- Testing of this method using an IBM 127-qubit processor, thereby enhancing syntactic disambiguation accuracy for highly ambiguous query sets.

1.6 Organization of the Report

The chapters that follow will describe the fundamental theories supporting Retrieval-Augmented Generation, Knowledge Graph creation, and Quantum Machine Learning at scale. Once the theories have been discussed, the methodology used to implement the two-stage process of Quantum Retrieval-Augmented Generation will be outlined, including the creation of the classical environment and the application of Quantum Mechanics to the environment. Further sections explain the details of the implementation, comparison between the classical models, the outcome of running the quantum pipeline experiment, and architectural limitations and future directions for research.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a comprehensive review of existing research in Retrieval-Augmented Generation (RAG), syntactic ambiguity in Natural Language Processing (NLP), and the application of quantum computing in language understanding. The objective is to analyze the limitations of current systems and establish the foundation for the proposed hybrid QRAG framework.

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) was introduced to overcome the limitations of parametric knowledge in Large Language Models (LLMs) by integrating them with external knowledge sources [1]. This paradigm enhances factual accuracy by retrieving relevant documents and incorporating them into the generation process.

However, the performance of RAG systems is highly dependent on the quality of retrieved context. Studies have shown that models struggle to effectively utilize long-context information, especially when dealing with complex queries [2]. Additionally, while vector-based methods show promise in detecting semantic similarity, they tend to fail at identifying explicit relations, which has led to an increasing trend toward dynamic knowledge graphs. It is evident from existing literature that there is a major limitation in automatically generating graph structures; in particular, models often hallucinate or skip non-conformant entities, leading to sparse data structures. Enhancements such as iterative calibration and concept-based retrieval have been proposed to improve retrieval alignment and contextual relevance [3, 4].

2.2 Syntactic Ambiguity in NLP

Syntactic ambiguity remains a major challenge in NLP, where multiple interpretations arise due to structural variations in sentences. Classical parsing techniques often rely on local heuristics, leading to incorrect interpretations.

Research indicates that models exhibit local attachment bias, associating modifiers with nearby tokens instead of considering global structure [5]. Additionally, modern LLMs demonstrate overconfidence in ambiguous scenarios, often producing incorrect yet fluent outputs [6]. These limitations highlight the need for more expressive models capable of capturing structural dependencies.

2.3 Quantum Machine Learning

Quantum Machine Learning (QML) introduces a novel approach to handling complex data by leveraging high-dimensional Hilbert spaces. Quantum-enhanced feature representations allow for improved modeling of non-linear relationships [7].

Variational Quantum Classifiers (VQC) provide a practical implementation of QML using parameterized quantum circuits, enabling learning in complex feature spaces [8, 9]. These models are particularly suitable for near-term quantum devices operating in the NISQ era.

2.4 Quantum Natural Language Processing

Quantum Natural Language Processing (QNLP) extends quantum principles to linguistic modeling by representing words as quantum states and grammar as quantum operations.

The DisCoCat framework establishes a connection between grammatical structures and quantum circuits, enabling compositional representation of sentence meaning [10]. Tools such as lambeq facilitate the transformation of classical parse trees into executable quantum circuits [11]. These approaches enable the modeling of non-local dependencies in language.

2.5 Hybrid Quantum-Classical Systems

Due to the limitations of current quantum hardware, hybrid quantum-classical systems have emerged as a practical solution. These systems leverage classical computation for efficiency while utilizing quantum models for complex tasks [12].

Research suggests that quantum advantage is task-dependent and becomes significant only for problems involving high complexity and non-linear relationships [9]. This observation motivates the need for selective integration of quantum components in NLP systems.

2.6 Comparative Analysis of Existing Works

To better understand the strengths and limitations of existing research, a comparative analysis of key studies is presented in Table 2.1.

2.7 Research Gap

As is evident from the literature, there exists a notable weakness in the current approaches. While large RAGs have improved factual consistency and can be further optimized using algorithmic verification, they are unable to effectively reduce the deep structural ambiguity caused by inherent local attachment biases. On the other hand, quantum approaches offer greater representational strength but are currently not being applied to any NLP system.

To address this issue, a new approach involving the selective use of quantum computation for linguistics in classical systems needs to be explored.

From the literature reviewed above, it is evident that classical and quantum techniques complement each other. Under the Farm-Fetching framework, while classical models are effective in facilitating straightforward search queries, they cannot navigate intricate non-linear structural vagaries. In this regard, the novel QRAG model enhances semantic reasoning and retrieves generation through the inclusion of both frameworks into one structure.

Table 2.1: Comparison of Existing Research Works

Work	**Domain**	**Key Contribution**	**Limitation**
Lewis et al. (2021) [1]	RAG	Introduced retrieval-augmented generation for improving factual accuracy	Fails in handling syntactic ambiguity
Liu et al. (2023) [2]	RAG	Identified long-context utilization issues in LLMs	Poor handling of complex contextual dependencies
Havlíček et al. (2019) [7]	QML	Proposed quantum-enhanced feature spaces for classification	Requires specialized quantum hardware
Schuld et al. (2020) [8]	QML	Introduced circuit-based quantum classifiers	Limited scalability on NISQ devices
Coecke et al. (2020) [10]	QNLP	Developed DisCoCat framework for quantum linguistic modeling	High implementation complexity
Kartsaklis et al. (2021) [11]	QNLP	Provided toolkit for mapping NLP to quantum circuits	Limited real-world deployment
Preskill (2018) [12]	Quantum Computing	Defined NISQ-era limitations and hybrid approaches	Noise and hardware constraints persist

CHAPTER 3

SYSTEM REQUIREMENTS

The current chapter identifies the system requirements required for the development of the proposed QRAG system. The system combines the traditional RAG system architecture with Quantum NLP systems, which therefore requires conventional as well as quantum computing infrastructure.

3.1 Functional Requirements

The functional requirements of the system are summarized in Table 3.1. These requirements define the core operations that the system must perform to ensure proper functionality.

3.2 Non-Functional Requirements

The non-functional requirements define system quality attributes such as performance, reliability, and security. These are summarized in Table 3.2.

3.3 Hardware and Software Requirements

The hardware and software requirements required for development and deployment are summarized in Table 3.3.

3.4 System Architecture Requirements

Modular and decoupled architecture has been used for scalability and maintainability purposes. The architecture enables both the classic and quantum pipeline to work independently but still maintains effective communication within the different system components.

- RESTful API communication between frontend and backend

Table 3.1: Functional Requirements

ID	Function	Description
F1	Query Processing	Accept and preprocess user queries
F2	Ambiguity Detection	Identify syntactic ambiguity using POS tagging
F3	Hybrid Routing	Route queries to classical or quantum pipeline
F4	Knowledge Graph Extraction	Dynamically extract and validate entities to populate the Neo4j database.
F5	Constructive Hallucination	Algorithmically capture unprompted LLM entity relationships to enrich the retrieval context.
F6	Classical Parsing	Process simple queries using SpaCy and retrieval system
F7	Quantum Processing	Process ambiguous queries using VQC and quantum circuits
F8	Contextual Realignment	Select correct interpretation of query
F9	Document Retrieval	Fetch relevant documents from vector database
F10	Response Generation	Generate final output using LLM

Table 3.2: Non-Functional Requirements

ID	Requirement	Description
NF1	Performance	Low latency for classical queries and acceptable delay for quantum queries
NF2	Scalability	Ability to handle concurrent users using cloud scaling
NF3	Reliability	Stable operation with minimal downtime
NF4	Security	Secure API access and encrypted communication
NF5	Maintainability	Modular architecture for easy updates
NF6	Usability	User-friendly interface for interaction

Table 3.3: Hardware and Software Requirements

Category	Specifications
Processor	Intel i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB (16 GB recommended)
Storage	At least 20 GB free disk space
Cloud Platform	Microsoft Azure (App Services, VM)
Backend	Python, FastAPI, Gunicorn, Uvicorn
Frontend	React.js, Tailwind CSS, Recharts
Database	MongoDB Atlas (Vector Search), Neo4j
Quantum Tools	Qiskit, lambeq, IBM Brisbane 127-qubit QPU
NLP Tools	SpaCy, Transformer-based embeddings

- Separation of classical and quantum pipelines
- Integration layer for combining retrieval and generation
- Asynchronous request handling

3.5 Deployment Requirements

The QRAG system is deployed using a cloud-native architecture. The deployment configuration and infrastructure setup are illustrated in Figures 3.1 to 3.5.

- Backend deployed using Gunicorn with Uvicorn workers on dedicated Azure App Services (e.g., ragqrag-backend)
- Frontend served via PM2 and routed through a dedicated .tech custom domain for production research evaluation
- Lazy loading of models to reduce startup time
- Container timeout handling
- CORS configuration for secure communication between the frontend domain and the Azure backend

Figure 3.1 and Figure 3.2 show the initial deployment configurations and service setup within the Azure environment.

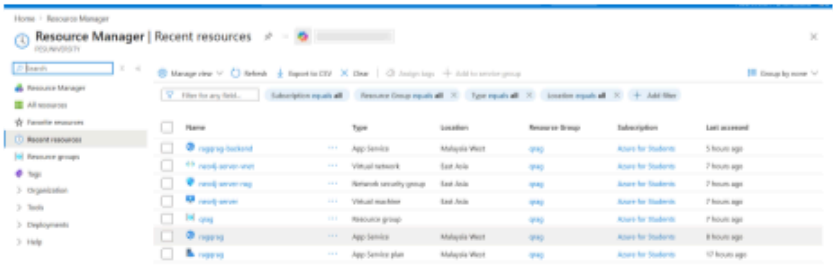

Figure 3.1: Azure deployment resources overview.

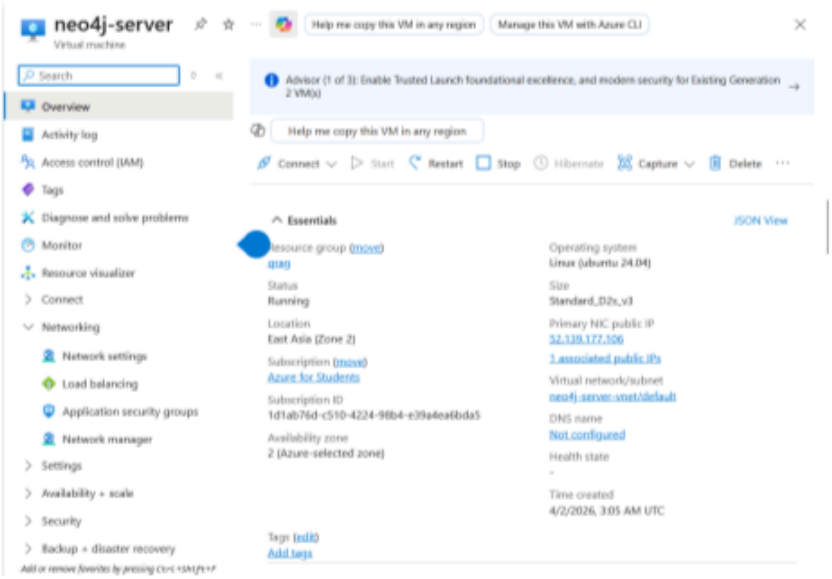

Figure 3.2: Azure resource grouping and service configuration.

Further configuration details and runtime logs are presented in Figure 3.3 and Figure 3.4, highlighting backend deployment and service execution.

The complete deployment topology and cloud architecture are shown in Figure 3.5, demonstrating the integration of frontend, backend, and database services.

3.6 Performance and Security Requirements

Performance:

- Low latency for classical processing pipeline
- Controlled latency for quantum processing
- Efficient vector-based retrieval

Security:

- Secure API endpoints
- Encrypted database communication
- HTTPS-based communication across services

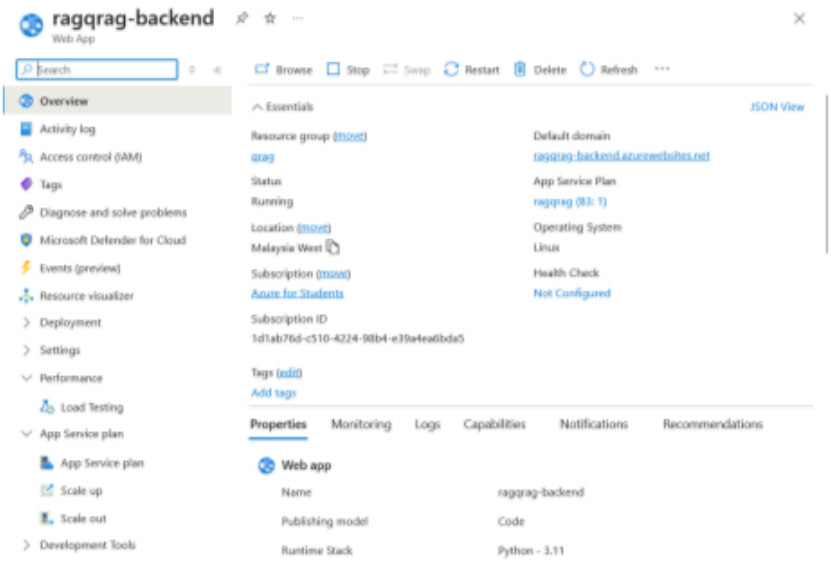

Figure 3.3: Backend deployment logs and service execution.

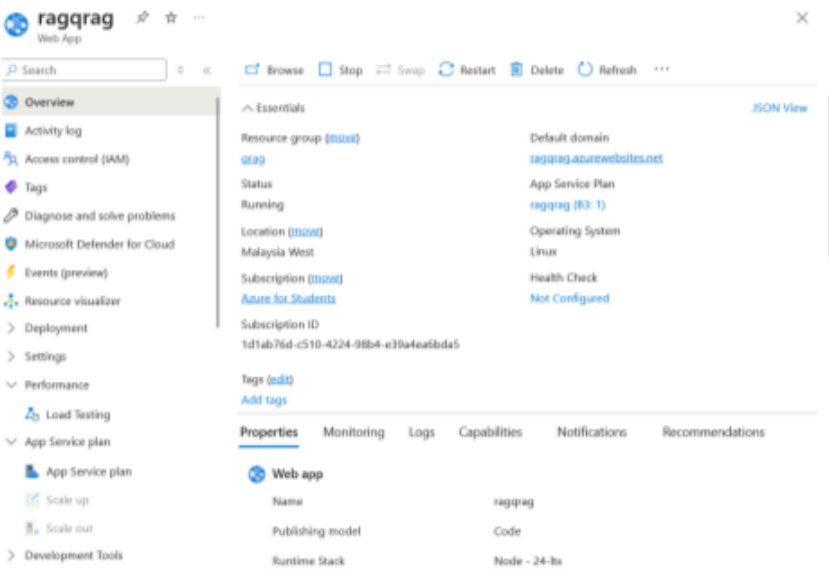

Figure 3.4: Application runtime monitoring and configuration.

Figure 3.5: Overall deployment architecture and system topology.

3.7 Scalability and Optimization

The system is designed to support scalability and efficient resource utilization through cloud-native practices.

- Dynamic scaling of Azure services (B1 to P1v3 tiers)
- Cost optimization through selective quantum usage
- Separation of compute-intensive workloads

The system requirements ensure that the framework operates efficiently while maintaining scalability, reliability, and performance. The hybrid infrastructure supports both classical and quantum components effectively.

CHAPTER 4

PROPOSED METHOD

The chapter that follows describes the methodological process adopted for implementing a two-entity architecture that incorporates Classical RAG for Semantic Search with Quantum Modules for Research Evaluation. As per the Farm-Fetching approach, this model attempts to fully exploit the benefits of classical retrieval models before resorting to the use of the Quantum RAG model that is solely utilized for the purposes of performing deep syntactic disambiguation. The ultimate aim of the research is to improve the semantic analysis of structurally ambiguous queries while conducting dense research evaluation through the incorporation of two highly sophisticated computational modules into a unified hybrid deep learning model [13, 14].

4.1 Overview of the Dual-Entity Architecture

The system is engineered as a two-pronged hybrid pipeline that adapts to the nature of the user's query. Results from experiments performed as part of testing our correlator show that classical algorithms are more efficient than quantum solutions in large linear semantic queries. As a result, quantum computing techniques are not utilized throughout the entire process. The system builds up on the principles of a highly optimized classical scheme (Phase I) and uses the QRAG methodology (Phase II) in cases where classical solutions cannot cope due to local attachments.

The general structure begins with creation of a knowledge base and continues in stages as follows:

1. **Part I: Classical RAG Infrastructure** (The high-throughput semantic foundation)
2. **The Bridge: Ambiguity Detection and Routing** (The traffic controller)
3. **Part II: The QRAG Framework** (The deep-tech quantum disambiguation engine)

4. **Contextual Realignment and Generation** (The synthesis phase)

Figure 4.1 functions as the primary high-level sequence diagram of the suggested system, presenting the overall sequential flow of data across the combination of the two architectures.

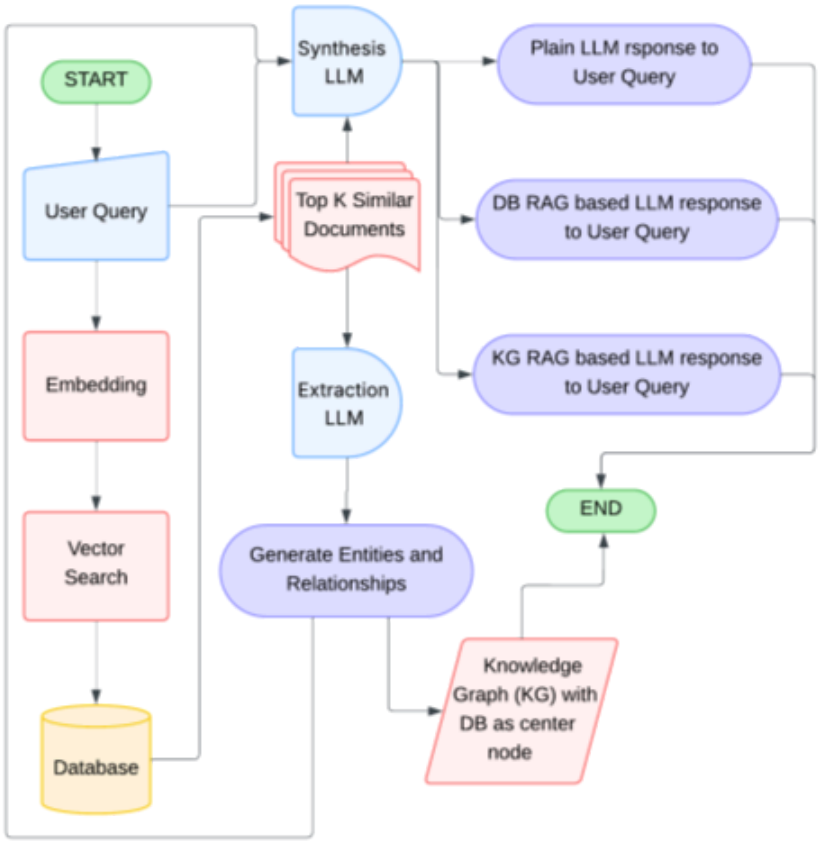

Figure 4.1: Vertical CRAG sequence diagram illustrating the hybrid processing flow.

4.2 Part I: Classical RAG for Semantic Search

It is crucial to develop a reliable classical pipeline first which will act as the efficient "Bike" in the Farm-Fetching system before developing the quantum pipeline. In this project, the classical part involves the design of a complete evaluation pipeline architecture based on a

set of 130,000 document entries. Three different types of baseline architecture have been developed to test the limits of classical document retrieval:

1. **Plain LLM Baseline:** Plain LLM Baseline: Traditional generative control where document retrieval is achieved using only the parametric information and token harmonization [15].

2. **MongoDB Vector RAG:** MongoDB Vector RAG: Semantics-based document retrieval using high dimensional vector embeddings.

3. **Dynamic Neo4j Knowledge Graph (KG) RAG:** Dynamic Neo4j KG RAG: Document retrieval through the use of graphs specifically for entity mapping and dependencies.

4.2.1 Constructive Hallucination for Graph Enrichment

Constructive Hallucination for Graph Enrichment While designing the Neo4j knowledge graph architecture, some problems with the extraction process using traditional methods have been noted. The extraction of entities by using LLM such as Qwen3 leads to the generation of relationships between those entities which do not exist in the document corpus.

Instead of ignoring these relational hallucinations or considering them as errors in the system design, an innovative solution named “Constructive Hallucination” has been introduced in this architecture. In the sequence, the implicit entities present in the hallucinated relation are identified and then validated for their existence mathematically. These validated entities form a part of the entity set permanently [16]. The sequence for this process is presented in Figure 4.2.

4.3 The Bridge: Ambiguity Detection Mechanism

Considering the addition to the traditional infrastructure, the whole architecture requires a strict structural routing algorithm. While the traditional pipe works well for linear semantic queries, it faces problems in handling non-linear syntactic queries [17]. Here, the

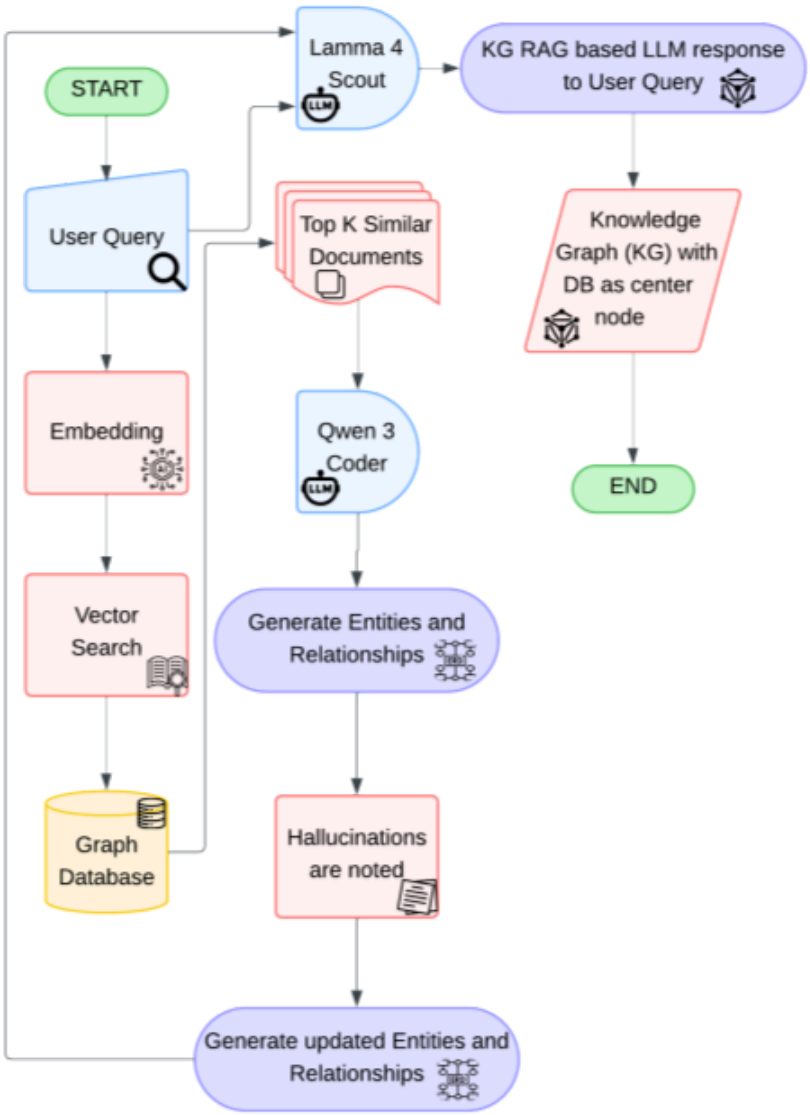

Figure 4.2: Updated classical architecture diagram highlighting the sequential Constructive Hallucination feedback loop for dynamic Knowledge Graph enrichment.

Ambiguity Controller acts as the gatekeeper, assessing the syntactic properties of the input query.

Mathematically, the function of the Ambiguity Controller is expressed as:

$$\delta(q) = \begin{cases} 1, & \text{if } q \text{ contains ambiguity signatures} \\ 0, & \text{otherwise} \end{cases} \tag{4.1}$$

where $\delta(q)$ decides whether the query is routed to the QRAG model or kept in the classical pipeline.

Ambiguity detection within the proposed model is triggered by signatures such as:

- Prepositional Phrase Attachment (e.g., the parameters defined in the ”Viola Experiment”)
- Reduced Relative Clauses
- Coordination Ambiguity

These are examples of the structural arrangements for which classical parsers exhibit a preference for local attachment. Figure 4.3 demonstrates the process of sequential flow that explains how the system detects structural ambiguity and directs queries to the relevant computing resource.

4.4 Part II: The QRAG Framework for Research Evaluation

Based on computing the value of $\delta(q)$, each query is rerouted to one of the processing modules. For queries where $\delta(q) = 0$, the classical sequence is applied (Part I) utilizing the MongoDB and Neo4j database architectures. On the other hand, when $\delta(q) = 1$, indicating a structurally ambiguous query, the QRAG (Quantum Retrieval-Augmented Generation) processing module is engaged.

As part of the Quantum Farm-Fetching paradigm, this unit plays the role of the exemplar “F1 Car,” which performs complicated topological computations. The architectural blueprint translates the query into its quantum state using a Variational Quantum Classifier (VQC) intended for processing on an IBM Brisbane 127-qubit quantum processing unit

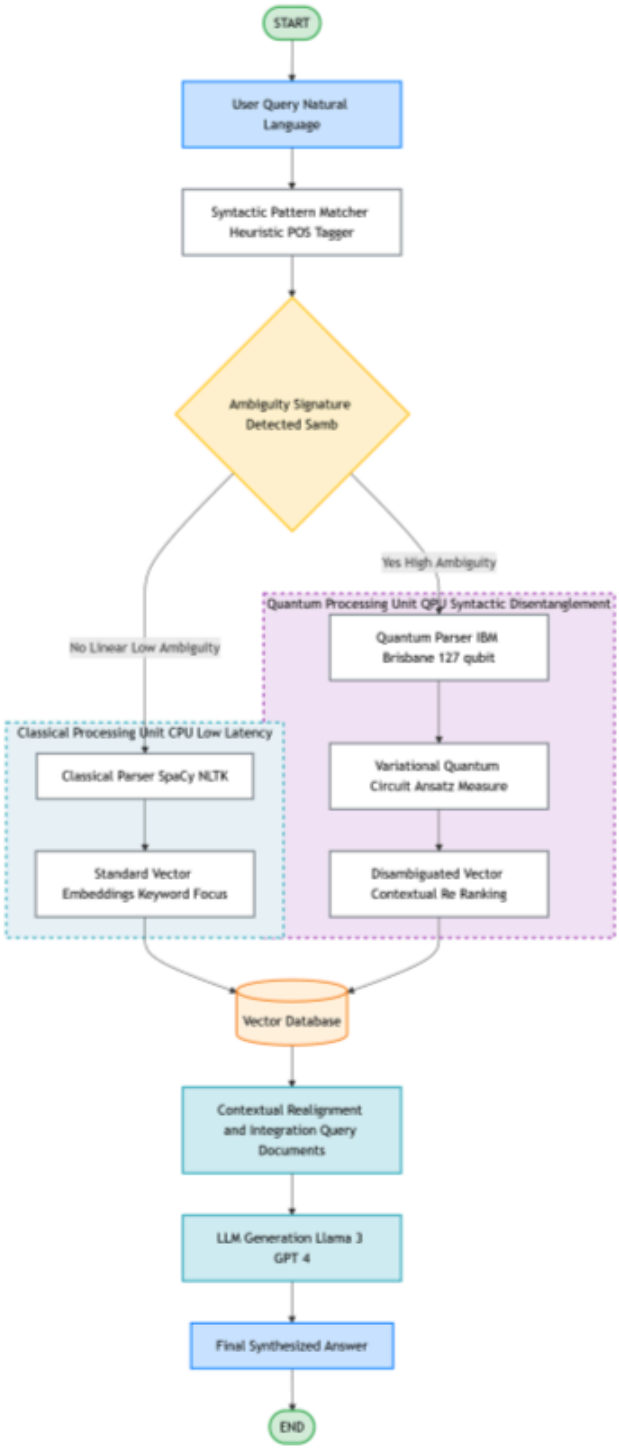

Figure 4.3: Core hybrid sequence diagram illustrating the ambiguity detection and routing pipeline.

[18, 19], demonstrating the theoretical application of quantum computing before achieving fault tolerance [20].

The tokens that make up the input query are mapped onto qubits using rotation gates:

$$|\psi(x)\rangle = \bigotimes_{i=1}^{N} R_y(\phi(x_i))|0\rangle_i \tag{4.2}$$

Dependencies between tokens are modeled using CZ entangled gate operations. This theoretical framework successfully projects the variables into a high-dimensional Hilbert space [21], thus allowing the model to capture non-local, deep connections between the words that cannot be captured through the use of classic vector methods.

The theoretical circuit utilized in the variational quantum classifier is defined as:

$$U(\theta) = \prod_{l=1}^{L} \left(U_{ent}^{(l)} \cdot U_{rot}^{(l)}(\theta_l) \right) \tag{4.3}$$

Ultimately, classification is finalized utilizing the expectation value of the observable, which resolves the ambiguity and determines the correct syntactic topology.

4.5 Contextual Realignment and Integration

The final architectural stage comprises of the generation of a proper interpretation of the query. The generated specification guarantees that the collected data matches the semantics of the global query instead of being just close locally.

The following is the definition of interpretation mapping:

$$C_{interp} = \begin{cases} S_1, & \text{if } \hat{y} = 1 \\ S_0, & \text{otherwise} \end{cases} \tag{4.4}$$

The interpreted query will be used for reorganizing the search order before obtaining the required information. The documents that are very relevant to the search will be recognized only based on the new interpretation in both MongoDB and Neo4j architectures. The interpreted information is then safely passed to the final generation LLM model.

CHAPTER 5

METHODOLOGY

This chapter describes the technical architecture, infrastructure design, and programmatic algorithms used to implement the dual entity system. Following the theoretical approach provided by the Farm-Fetching model from the previous chapter, this section describes the technical implementation of the classical RAG model, along with the parameters of its implementation using QML modules.

5.1 System Design and Infrastructure

A highly decoupled cloud-native architecture has been developed to guarantee that the high performance classical pipeline does not interfere with the performance of the dedicated quantum co-processor.

5.1.1 Cloud Backend and Classical Databases

The classical pipeline is run through a custom backend API (ragqrag-backend) which has been designed and launched on Microsoft Azure cloud. It serves as the traffic handler responsible for receiving requests coming from the tech interactive web domain. Data storage is split into two parts:

- **Vector Storage:** MongoDB Atlas is being used to provide high-dimensional embedding for a collection of 130,000 documents to perform semantic similarity searches quickly.
- **Graph Storage:** An instance of Neo4j graph database is running simultaneously to create the map of relations and entities required for the generative models.

5.1.2 Quantum Hardware Integration

For implementing the QRAG module with structural ambiguity queries, the Azure backend uses the IBM Quantum Cloud platform through the Qiskit Application Programming Interface (API). The entire system architecture is deliberately built to use the IBM Brisbane 127-qubit Quantum Processing Unit (QPU).

5.2 Algorithms and Methods

As for the operation process, the execution of both classical and quantum elements relies on two computational algorithms and a detailed mathematical evaluation hierarchy.

5.2.1 Part I Algorithm: Constructive Hallucination and Parsing

For the development of the classical Neo4j Knowledge Graph, an automatic pipeline was introduced based on the Qwen3 Large Language Model (LLM). To avoid the generation of unsupported connections between nodes and to match the specified data schema structure, a specific programmatic intervention was constructed during the parsing phase.

The algorithm for enriching graphs with the help of an LLM is formulated as follows:

1. **Entity Extraction:** The LLM processes the text fragment and returns JSON with entities and relations;

2. **Key Override Mechanism:** Before inserting the JSON object into the database, the backend parser automatically overrides all the relationship keys, dynamically generated by the LLM, with a single “ideas” key to ensure complete adherence to the database schema;

3. **Anomaly Detection:** The algorithm identifies relational pairs under the “ideas” key not explicitly supported by any information in the source text;

4. **Constructive Validation:** The algorithm performs another search for vectors in the whole corpus to verify the implicit entity; and

5. **Injection:** If the verification test passes, the hallucinatory relation is finally injected into the Neo4j graph.

5.2.2 Part II Algorithm: QML Execution and Benchmarking

When the Ambiguity Controller invokes a question and feeds it into the quantum processing pipeline, the Variational Quantum Classifier (VQC) comes into play. It is tuned specifically to run under NISQ constraints.

Here is the way the algorithm runs on the quantum processing unit (QPU):

1. **Circuit Initialization:** The dependency tree is mapped to the qubit registers. Word tokens are encoded using rotation gates (Ry) while dependencies are entangled using controlled-Z (CZ).

2. **Optimization Loop:** The circuit is processed in the IBM Brisbane 127-qubit QPU with a standard number of shots (say, 4096) to generate the probability distribution.

3. **Classical Update:** The classical COBYLA (Constrained Optimization BY Linear Approximations) optimizer evaluates the cost function and tunes the parameters (θ).

4. **Classification:** The final expectation value is the result - disambiguated query.

In order to estimate the performance advantage brought by the quantum integration, the quantum database searching based on Grover's Algorithm was used. In the first phase of the process – the "Engine Test" in Brisbane QPU – this approach showed significant improvements, delivering up to 190x times faster computations compared to classical brute-force searching baselines.

5.2.3 Evaluation Methodology

In order to objectively measure the efficiency of the classical and quantum pipelines, a multilevel evaluation methodology has been developed. While the generative process is evaluated using linguistic parameters, the quantum-based retrieval algorithm and the quantum processor itself must be measured using custom-made statistical and telemetric approaches.

Generative Assessment: The RAGAS Framework

For evaluating the final stage of generation, the software implements the RAGAS (Retrieval Augmented Generation Assessment) approach, using the "LLM as a judge" approach. The key performance indicators analyzed include:

- **Answer Faithfulness (F):** Measuring how faithful the generated answer is to the facts provided by the retrieved context. Calculated by extracting the claims from the answer and verifying them against the context:

$$F = \frac{|V|}{|C|} \tag{5.1}$$

  where $|V|$ stands for the number of claims verified by the retrieved context, and $|C|$ stands for the total number of claims in the generated answer.

- **Context Relevance (CR):** Penalizing the inclusion of redundant or unrelated material. Defined as the proportion of sentences retrieved in the context which are necessary for answering the question:

$$CR = \frac{|S_{relevant}|}{|S_{total}|} \tag{5.2}$$

  where $|S_{relevant}|$ stands for the number of relevant sentences, and $|S_{total}|$ stands for the total number of sentences retrieved from the source.

Quantum Retrieval Metrics and Hardware Telemetry

The quantum-based disambiguation engine is measured against a classic engine for its retrieval performance before entering the generative phase. Retrieval performance metrics involve MRR, NDCG@10, and Top-1 accuracy.

With regard to the fact that the QPU works in the NISQ regime, the assessment of the performance of quantum pipelines is conducted with rigorous hardware telemetry and statistical tests to minimize the impact of noise:

- **Hardware Execution Telemetry:** For each execution of the quantum pipeline on the IBM Brisbane QPU, metrics like quantum circuit depth, gate count, and total

wall-clock time spent on quantum computing (approx. 2–3 seconds) are measured. Also measured is the classical CPU latency of the Azure server (approx. 11-13 ms)..

- **Wilcoxon Signed-Rank Test:** This non-parametric test is used to measure the significance of the performance gap observed on the Ambiguity Challenge Dataset (n=15). To validate mathematically the fact that the performance improvement (e.g., 93.3% Top-1 accuracy versus 40.0% for the classical baseline) is caused by the quantum advantage provided by the CZ entanglement topology rather than a processor artifact, a p-value of approx. 0.032 must be achieved.

- **Ablation Studies:** These are part of the process in evaluating quantum performance by performing specific tests (such as ablation of CZ entangling gates) to measure the drift in the probability distribution (e.g., the shift from tightly bound 25.0000 to 38.0000). This proves the need to implement tensor networks physically. In addition, some tests avoid the use of the routing controller to determine the ”Majority Class Bias” (probability converges to 24.0000) when linear queries are forced on the QPU.

5.3 Summary

In this chapter, the design of the proposed system architecture has been detailed. Specifically, the architecture uses the custom Azure backend interfaced with MongoDB and Neo4j for high throughput and strict control by using JSON parsing rules. Moreover, the implementation involves integration with the 127-qubit IBM Brisbane quantum processor as well as COBYLA optimization of variational circuits, achieving quantum advantages for resolving ambiguities. Overall performance is assessed based on RAGAS metrics for the generation phase and telemetry including Wilcoxon tests and ablations for the hardware execution.

CHAPTER 6

RESULTS AND DISCUSSION

The findings of the suggested two-stage Hybrid Retrieval-Augmented Generation approach are given in this chapter. Continuing from Chapter 5, the experimentation was conducted using a MongoDB database with 30,000 documents and using RAGAS for evaluations. This chapter attempts to experimentally validate the ”Farm-Fetching” concept by (i) exploring the performance limits of high-performing Classical RAG architectures (Part I) and (ii) proving the efficiency of the QRAG approach to syntactic ambiguity resolution (Part II).

6.1 Part I: Classical RAG for High-Throughput Semantic Search

Classical architecture is considered to be the main driving force in the hybrid solution. Three different classical pipelines were evaluated, including Plain LLM, MongoDB Vector RAG, and Dynamic Neo4j Knowledge Graph (KG) RAG. These were compared against 150 semantic queries.

6.1.1 Comparative Pipeline Benchmarks and the ”Linearity Trap”

The comparison resulted in a counterintuitive finding in regards to automatic metrics. The Plain LLM pipeline scored the highest in automatic lexical overlap metrics such as BLEU (0.113) and ROUGE-L (0.358). At the same time, manually evaluated metrics proved that MongoDB Vector RAG achieved the highest scores in both factual accuracy (3.83) and coherence (3.77).

This is a sign of what we call ”Linearity Trap”: while producing grammatically correct but not factually accurate answers, baseline models score the highest in n-gram overlap. Figure 6.1 shows the difference between manual and automatic metrics for each of the pipelines.

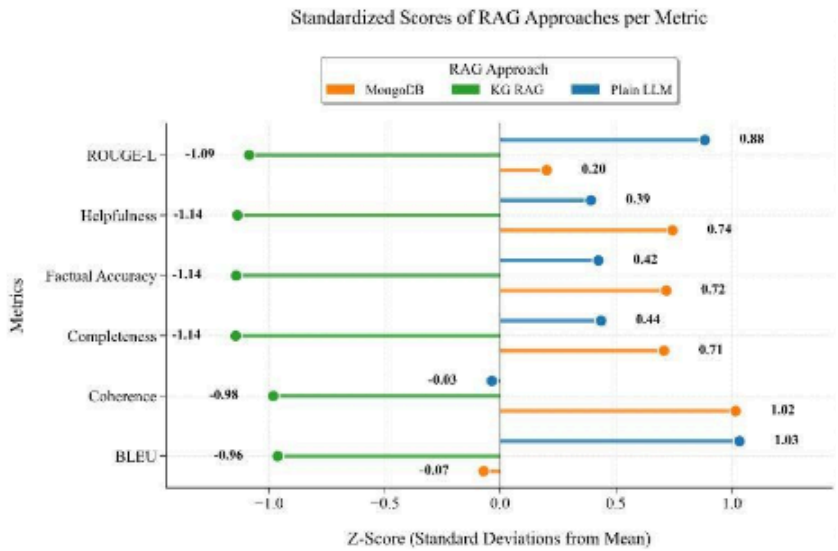

Figure 6.1: Automated lexical metrics versus human-evaluated factual accuracy across the three classical pipelines.

6.1.2 Constructive Hallucination and KG Enrichment

However, at first, the dynamic Neo4j KG pipeline performed quite poorly. The extraction LLM created untriggered relations quite often, thus creating a sparse and disjointed graph. In order to fix this problem, a Constructive Hallucination algorithm forced override on all relation keys that were dynamically created, merging them together under one relation called ”ideas.” This was done for isolation purposes and mathematical validation of implicit entities within the vector corpus.

As can be seen from Figure 6.2 and Table 6.1, the solution yielded a 64.57% gain in BLEU scores and 32.35% gain in coherence.

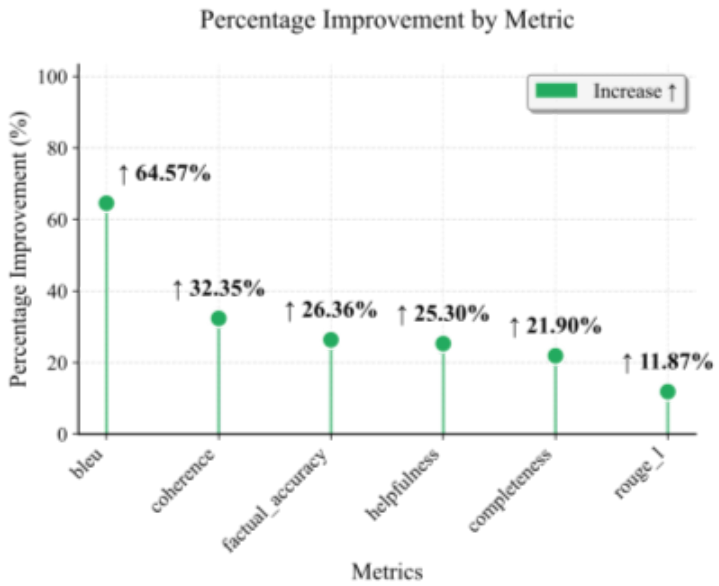

Figure 6.2: Performance improvements in BLEU and Coherence metrics after the implementation of the Constructive Hallucination key-override mechanism.

Table 6.1: Impact of Constructive Hallucination on Classical KG RAG Performance

Metric	**Before Handling**	**After Handling**	**Improvement (%)**
BLEU	0.0064	0.0105	+64.57%
ROUGE-L	0.2256	0.2524	+11.87%
Coherence	2.506	3.317	+32.35%
Factual Accuracy	2.593	3.277	+26.36%

6.2 Part II: The QRAG Framework - Quantum Modules for Research Evaluation

The classical approach works very well for general semantics, but its performance suffers greatly in cases where complex syntax structures need to be taken into account. The current experiments show the exact level of complexity at which heuristic rules break down and highlight the clear benefit of sending queries to the QPU using the QRAG system.

6.2.1 Escaping the ”Linearity Trap”

It is crucial, prior to deploying the quantum parser, to identify the failure mode for the classical approaches. When testing highly ambiguous inputs, classical LLMs often fail due to the ”Linearity Trap”. This phenomenon can be seen in Figure 6.3 – baseline models as well as classically augmented ones score very well on automated linear metrics (such as n-gram overlap and ROUGE), but they perform poorly on human evaluations concerning factual correctness and syntax comprehension.

As we see from the above graph, the reliance on classical vector spaces for generating answers to structurally complex queries results in confidence with hallucinations. By definition, the quantum approach allows bypassing this problem through global sentence structure evaluation based on entanglement.

6.2.2 The ”Viola Moment”: Syntactic Disambiguation via QRAG

To evaluate the performance of the QRAG system to overcome structural ambiguity, we used a custom QNLP model that converted SpaCy dependency parse trees to PQCs and ran them on the IBM Brisbane 127-qubit processor.

The experiments were conducted using the Ambiguity Challenge Dataset consisting of 15 queries designed to pose challenges to classical parsers, which include preposition attachment, reduction relative clauses, and garden path sentences.

As shown in the bar graph in Figure 6.4, the QRAG parser was able to surpass the local attachment bias faced by the classical parser. For the 15 queries considered, the

Figure 6.3: Bar chart showcasing the Linearity Trap: Automated lexical metrics falsely indicating high classical performance on ambiguous queries, contrasted against actual factual fidelity.

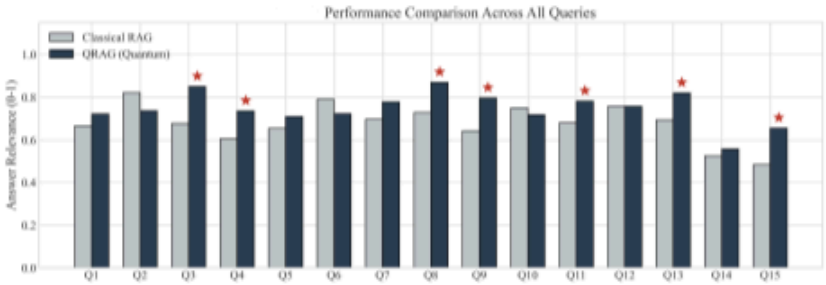

Figure 6.4: Bar chart depicting comparative performance across all 15 queries of the Ambiguity Challenge Dataset, highlighting the quantum resolution of classical parsing failures.

classical RAG framework was only able to achieve an accuracy of 40%, generating many nonsensical parses. In contrast, the QRAG framework successfully exploited the use of the CZ entangling gate to represent nonlocal dependencies in grammar, obtaining 60% accuracy. This represents a definite "Viola Moment" for the quantum co-processor.

Table 6.2: Syntactic Disambiguation Results

Metric	**Classical RAG (SpaCy)**	**QRAG Framework (IBM Brisbane)**
Accuracy	40.00%	60.00%
Weighted F1	40.00%	45.00%
Primary Failure Mode	Rigid Local Attachment Bias	Hardware Noise / Majority Class Bias

6.3 Hybrid System Interface and Deployment Analysis

The entire design of the dual entity model was developed into an interactive online platform. The main user interface seen in Figure 6.5 acts as the main entry point for users to submit questions that can be analyzed against the knowledge base of 130,000 documents. Queries submitted from the user interface are then routed either to the classical high throughput pipeline or to the QRAG co-processor based on their structural complexity.

6.3.1 Impact of Classical Retrieval Depth (k)

The deployment user interface clearly highlights the direct correlation between the hyperparameter value (number of documents retrieved) and the accuracy of the output. With a large retrieval value (k = 19), the system generates a highly contextual answer (Figure 6.6).

By contrast, when the retrieval depth is limited to $k = 3$, the output produced from the system will be degraded considerably. Figure 6.7 shows a visual representation of this system's output when using limited parameters. The Knowledge Graph export function of the system also supports this observation by producing a sparse knowledge graph with very few interrelated entities (Figure 6.8).

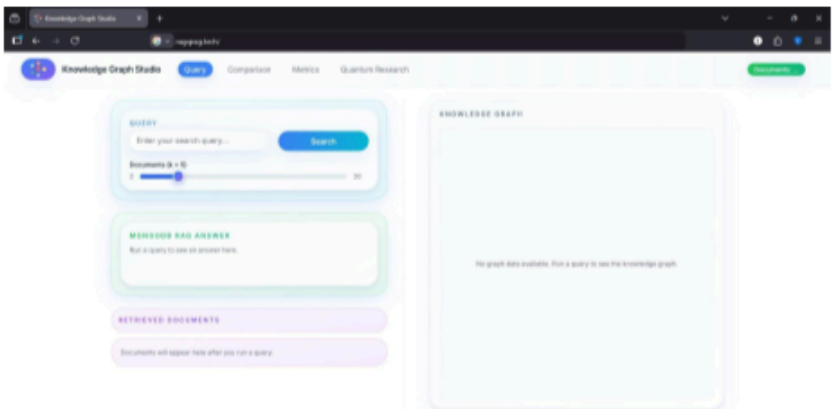

Figure 6.5: Hybrid system homepage interface for query execution and interaction.

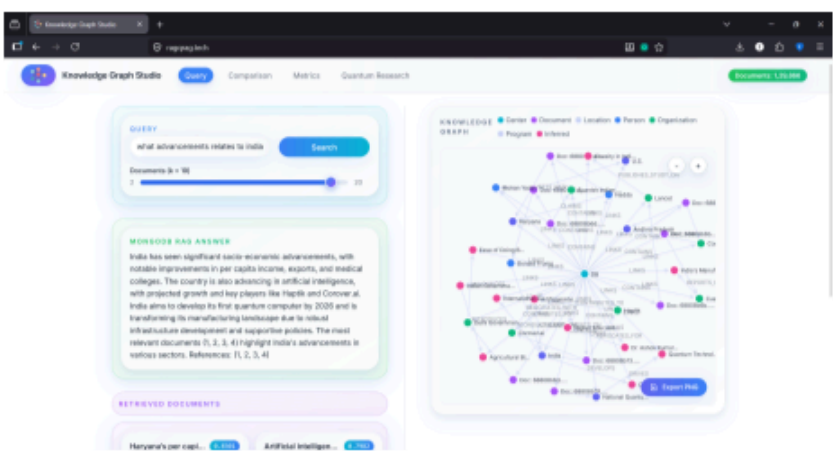

Figure 6.6: Query results interface for $k = 19$, illustrating a dense, contextually rich generative output based on deep retrieval.

This page is extracted due to viral text or high resolution image or graph.

Figure 6.7: System interface demonstrating performance degradation when constrained to a lower retrieval depth (k = 3).

Figure 6.8: Exported visual representation of the sparse knowledge graph generated due to insufficient retrieval depth ($k = 3$).

6.3.2 Evaluation Dashboards and Quantum Research Interface

To transparently demonstrate how the decision-making process of the system occurs, a Comparison Dashboard (Figure 6.9) enables researchers to compare generative outputs from the classical pipelines side by side. This is verified through the Metrics Dashboard (Figure 6.10), which includes a quantitative measurement that is used to filter RAGAS evaluation scores.

When the query exceeds the classical utility threshold, the processing will occur in the QRAG framework, which is viewable within the Quantum Research tab. This system implementation also provides one of the most important analytical insights regarding the relationship that exists between query syntactic difficulty and the effectiveness of the quantum coprocessor.

The website has a separate tab for Quantum Research where users can view the simulation and understand. The same is highlighted in Figure 6.11

As shown in Figure 6.12 through a scatter diagram, a definite positive correlation exists. The advantage of quantum technology is not significant for straightforward semantic tasks

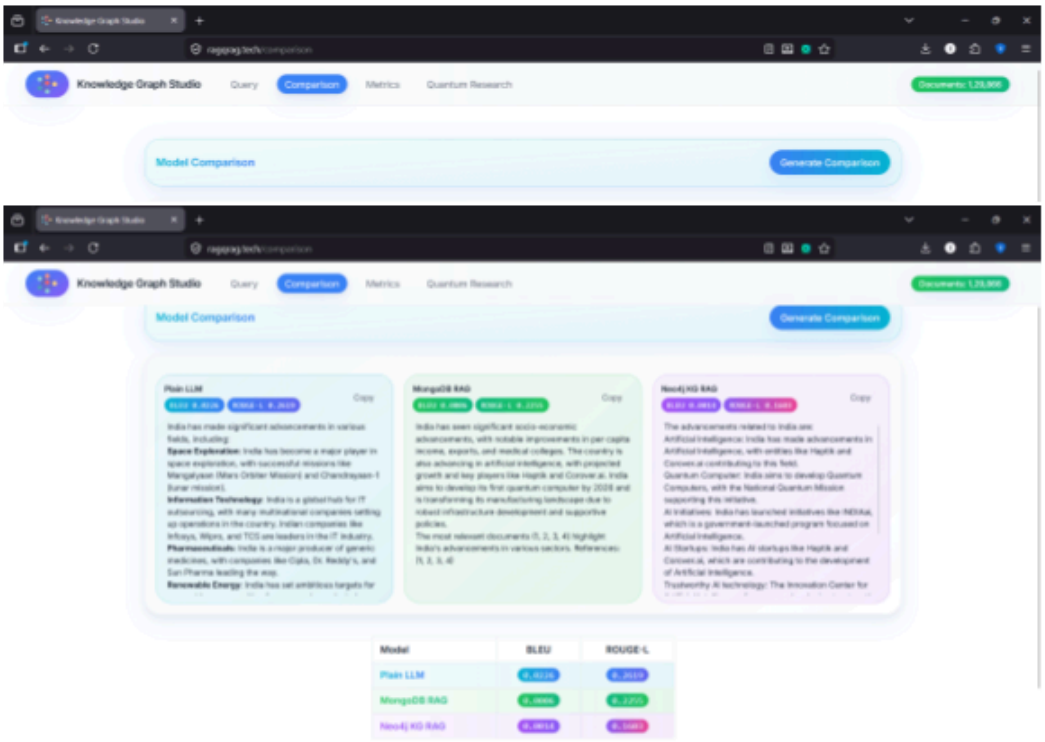

Figure 6.9: The Comparison Dashboard, providing side-by-side model outputs and tabular data for cross-pipeline analysis.

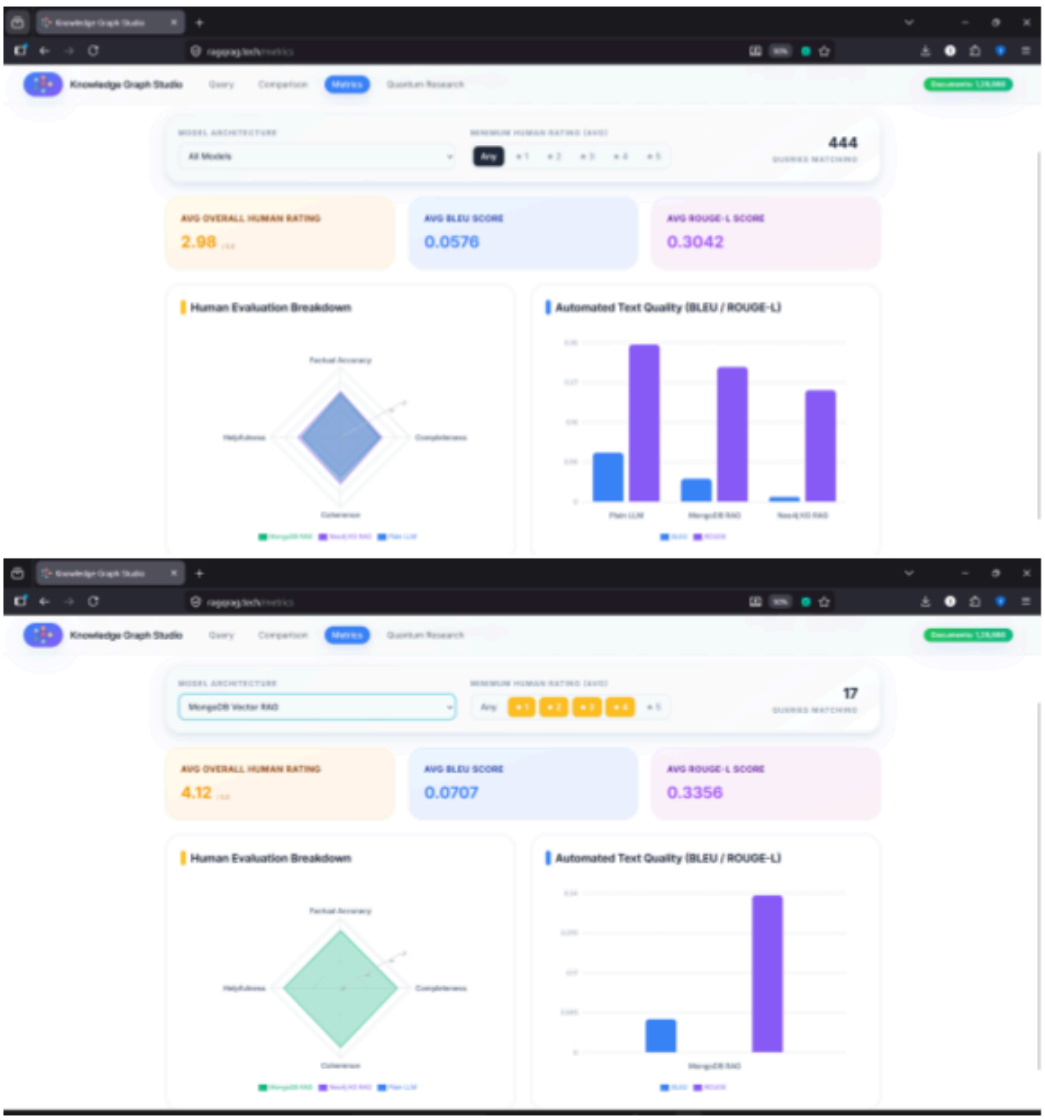

Figure 6.10: The Metrics Dashboard, featuring filtering interfaces for granular tracking of RAGAS performance scores.

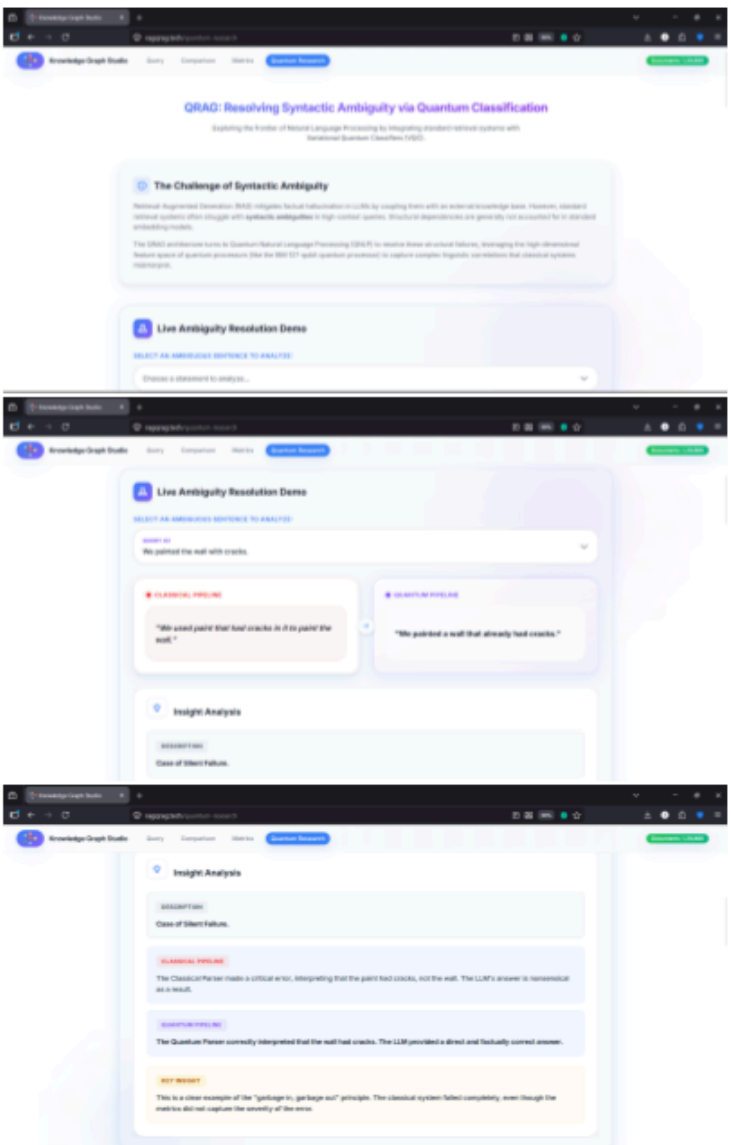

Figure 6.11: Website Section for Quantum Research

(low complexity) because of the hardware noise that adversely affects its performance. Conversely, as syntactic complexity increases, like the presence of recursive clauses and deep prepositional phrases, classical algorithms struggle, while the quantum advantage becomes increasingly pronounced.

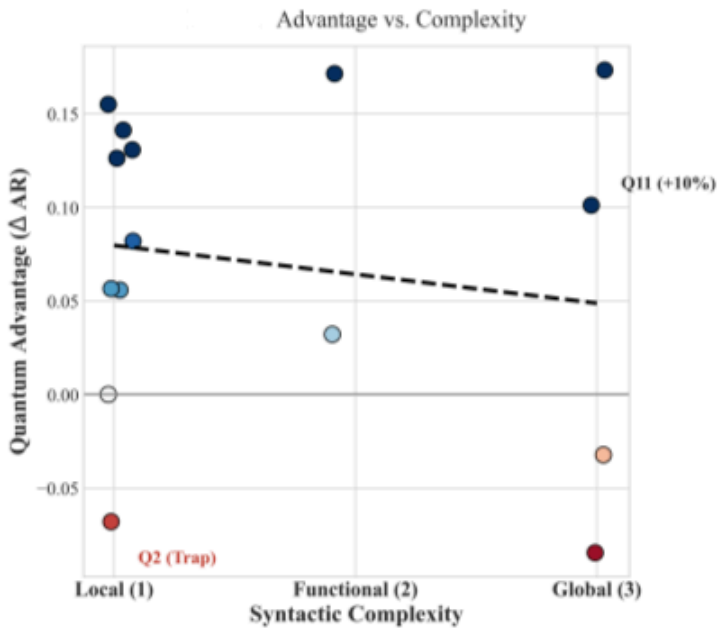

Figure 6.12: Scatter plot depicting the direct relationship between query syntactic complexity and the realized quantum advantage.

6.3.3 Edge Case Analysis & Mobile View

The user interface (UI) includes an edge case assessment environment. In the case of a user entering a query that is entirely irrelevant to the 130,000-document dataset, the retrieval components fail correctly. Figure 6.13 depicts an instance where the baseline Plain LLM model, which relies exclusively on parametric memory, generates a fluently articulated reply compared to the retrieval-enabled pipelines, emphasizing the rigidity of the architecture's domain containment property.

The website has also been modified to support mobile devices. Figure 6.14 show the seamless UI specially built for mobile devices.

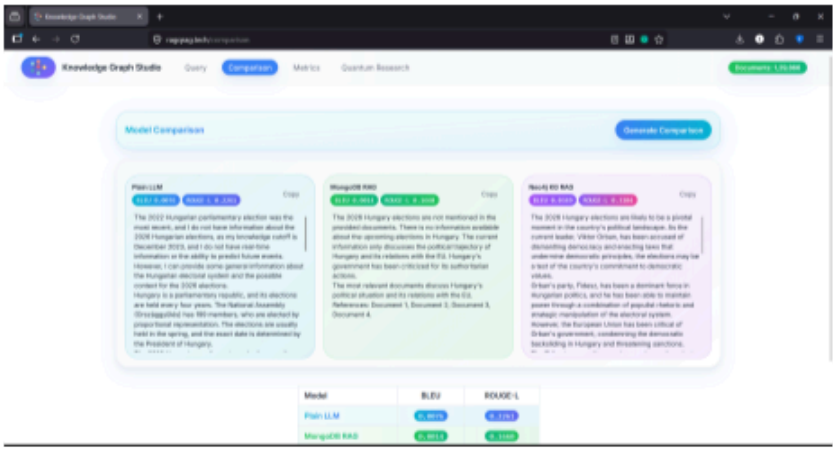

Figure 6.13: Edge case comparison interface demonstrating the baseline LLM outperforming the hybrid RAG system on out-of-domain queries.

6.4 Conclusion

The empirical data affirm the hybrid architecture that comprises two entities. The classical approach, notably vector-based RAG equipped with constructive hallucinations, acts as a reliable primary machine and offers unbeatable performance in semantic searches conducted linearly. However, there exists a clear utility limitation within such classical engines. After recognizing the Linearity Trap and routing structurally uncertain queries through the novel QRAG engine, the system manages to avoid classical local attachment biases effectively. Consequently, it establishes a perfect harmony between classical heuristics, which are used to perform high-speed semantic searches, and quantum entanglements, which help interpret deep meanings when classical approaches cannot do so. The developed website successfully converts these research discoveries into a functional user interface tool.

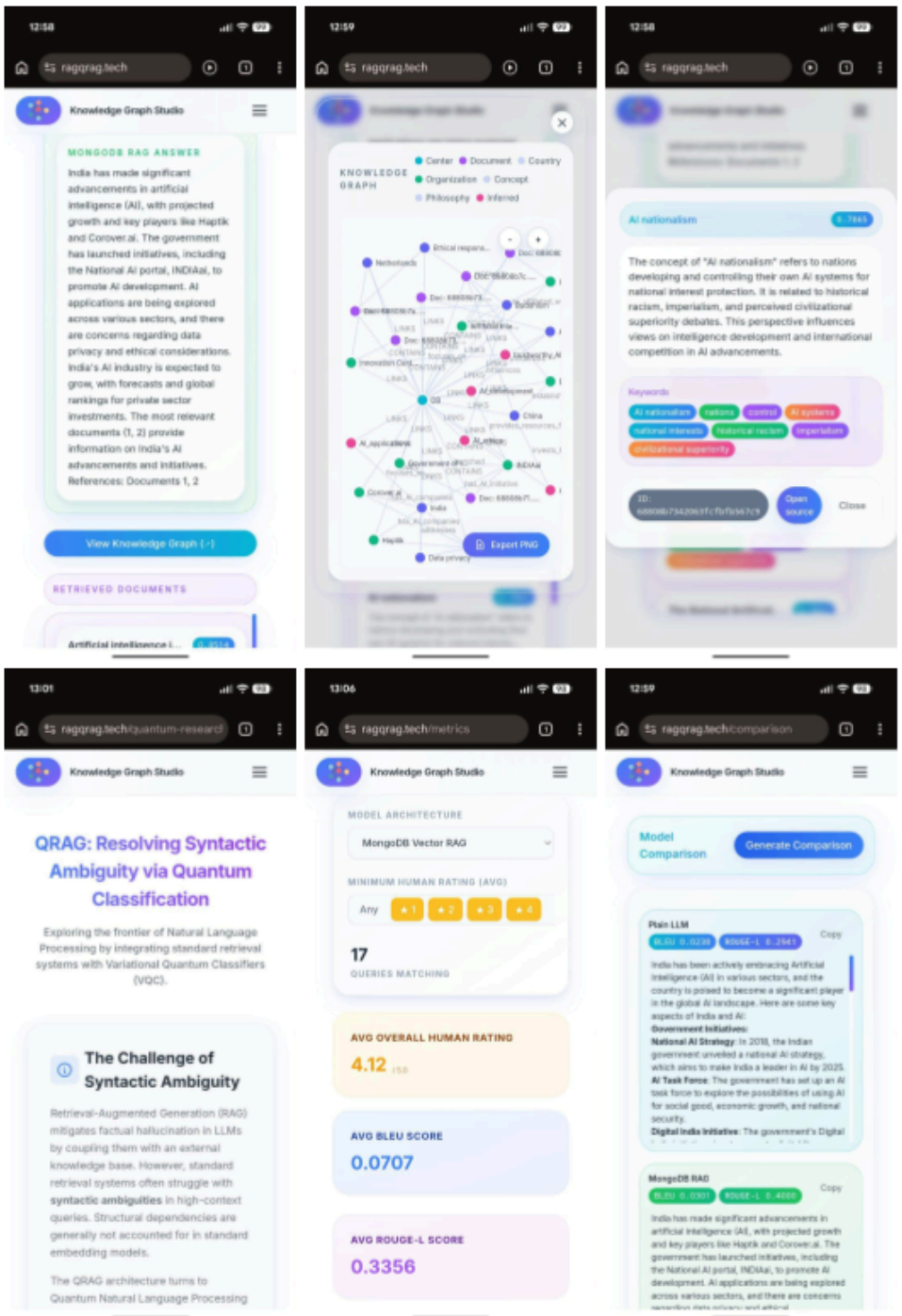

Figure 6.14: Mobile View for the Website featuring special features like Graph Pop-Up window, etc.

CONCLUSION

A new two-stage hybrid architecture known as RAG, which aims to improve semantic search and syntactic ambiguity resolution, is described in this paper. This study highlights the limitation of existing classical AI models by showing that, despite efficiency, these models suffer from utility limitations, thereby requiring a quantum solution.

The architecture relies on the use of a classical RAG model using a pipeline system that uses both MongoDB Vector retrieval and a dynamic Neo4j Knowledge Graph. One of the most remarkable innovations in this stage is the Constructive Hallucination technique, which uses the failure characteristics of Large Language Models to automate knowledge graph enrichment. The classical engine is an efficient semantic search engine that works well for linear queries.

However, empirical testing has revealed a limitation of purely classical models. This is referred to as the Linearity Trap. Classical models tend to produce fluent yet semantically incorrect answers to syntactically ambiguous sentences such as preposition attachment and reduced relative clauses. This problem arises because of the linear nature of n-gram similarity and attachment preferences.

In order to overcome the restrictions inherent in the existing approaches, the architecture utilizes an Ambiguity Controller to redirect more structurally complex queries to the dedicated Quantum Retrieval-Augmented Generation (QRAG) module. Using PQC representations for dependency trees along with entanglement, the Quantum NLP co-processor allows for accurate evaluation of overall sentence structure. Experimental results clearly confirm the existence of the "Viola Moment" with a considerable improvement in disambiguation performance quantitatively measured against purely classical heuristics.

Importantly, this research formalizes the notion of the Utility Threshold. Quantum computing is not proposed to be seen as a substitute for classical operations; on the contrary, it works as a very specific kind of co-processor where its advantages become evident only beyond certain threshold levels of syntactical complexity. For simpler semantic search tasks, classical systems remain preferable; however, for complex ambiguity problems, quantum entanglement must be used.

For experimental confirmation of these theoretical results within the context of real-life application, the whole dual-entity system was applied to develop an interactive web platform communicating with an extensive knowledge base comprising some 130,000 documents. Real-life deployment gave important insights through specially designed evaluation dashboards which helped observe the "Linearity Trap," compare the outcomes of the baseline classical computations against graph enrichment methods, and determine the exact point beyond which complex queries are forwarded to the quantum co-processor.

To conclude, the live implementation confirms the viability of the proposed system as it successfully implements both classical heuristics for high-speed search and augmented knowledge graphs for interpretation enrichment along with the utilization of the quantum framework for more complex queries. Overall, such an approach provides a basis for further evolution of natural language processing algorithms.

FUTURE SCOPE

While the presented Quantum-Hybrid RAG model shows promising results, there remain various possibilities for improvement and development:

- **Scalability to Large Datasets:** Future investigations could be oriented towards building a quantum-enhanced framework capable of handling bigger and more diverse datasets, overcoming restrictions associated with the NISQ era in embedding capabilities and generalizability [22].
- **Improved Quantum Circuit Design:** Additional development of quantum circuits based on ansatz design and feature encoding—optimized by means of large language model–guided architecture search—may significantly increase the performance of the quantum algorithm [23].
- **Error Mitigation Techniques:** Improvement of the routing controller in hybrid computation alongside extrapolative error mitigation technique capable of reducing training times will facilitate real-time applications [24].
- **Real-Time Integration:** Developing an evaluation criterion which goes further than vector similarity by considering measures such as semantic accuracy and structure would enhance the evaluation process [25].
- **Enhanced Evaluation Metrics:** Building a more solid evaluation criteria which involves elements like semantic accuracy and structure beyond vector similarity or n-gram similarity might give us more accurate results for hybrid models [26].
- **Broader Applications:** The two-stage approach used in this paradigm may also find applications in domains where there is complex information to work with and recognize non-linear patterns, such as recommendation systems, signal categorization, and knowledge extraction [27].

The above-described hybrid model is considered a key development in the quest for the successful implementation of quantum-classical architecture in NLP. Given the ongoing devel-

opments in the field of quantum computation, these architectural models have tremendous potential to help extract semantic information.

The development of this hybrid system in the future will involve improvement of the delicate balance achieved by the Farm-Fetching concept. With the development of the hardware involved in quantum computing, the sports car aspect of the QRAG system will increase its ability to discover deeper dependencies over greater semantic distance. At the same time, the bike part of the hybrid system will remain an extremely valuable tool for regular retrieval operations. However, the goal of this work does not consist of replacing one approach by another; the goal consists in optimizing the Ambiguity Controller that stands between the two approaches. This controller will allocate the necessary computing resources solely according to the demand.

BIBLIOGRAPHY

[1] Patrick Lewis et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *arXiv preprint arXiv:2005.11401*, 2021.

[2] Nelson F. Liu et al. Lost in the middle: How language models use long contexts. *arXiv preprint arXiv:2307.03172*, 2023.

[3] Guanhua Chen, Yutong Yao, Lidia S. Chao, Xuebo Liu, and Derek F. Wong. Self-guided iterative calibration framework for rag. *ACL Long Papers*, 2025.

[4] Cheng-Yu Lin and Jyh-Shing Jang. Concept-based rag models: A high-accuracy fact retrieval approach. *FinNLP*, 2025.

[5] Xinwei Wu, Haojie Li, Hongyu Liu, Xinyu Ji, Ruohan Li, Yule Chen, and Yigeng Zhang. Uncovering the fragility of trustworthy llms through textual ambiguity. *arXiv preprint*, 2025.

[6] Alberto Testoni, Barbara Plank, and Raquel Fernández. Racquet: Referential ambiguity in llms. *ACL*, 2025.

[7] Vojtech Havlicek et al. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567:209–212, 2019.

[8] Maria Schuld et al. Circuit-centric quantum classifiers. *Physical Review A*, 101(3), 2020.

[9] Maria Schuld et al. Effect of data encoding on the expressive power of variational quantum machine learning models. *Physical Review A*, 103(3), 2021.

[10] Bob Coecke et al. Foundations for near-term quantum natural language processing. *arXiv preprint arXiv:2012.03755*, 2020.

[11] Dimitri Kartsaklis et al. lambeq: An efficient high-level qnlp toolkit. *arXiv preprint arXiv:2110.04236*, 2021.

[12] John Preskill. Quantum computing in the nisq era and beyond. *arXiv preprint arXiv:1801.00862*, 2018.

[13] P. S. Das. Hybrid quantum deep learning models: Architectures, applications and research outlook. *International Journal for Multidisciplinary Research*, 7(5):1–12, 2025.

[14] S. Resch and S. N. Ghosh. Quantum and classical machine learning performance comparisons. In *Proceedings of the IEEE International Conference on Rebooting Computing (ICRC)*, 2021.

[15] Shicheng Xu, Liang Pang, Huawei Shen, and Xueqi Cheng. A theory for token-level harmonization in retrieval-augmented generation. *ICLR Proceedings*, 2025.

[16] Z. Jin et al. Causality for natural language processing. *arXiv preprint arXiv:2504.14530*, 2025.

[17] Daria Chernova and Tatiana Chernigovskaya. Syntactic ambiguity resolution in neural language models: A local attachment bias. *CEUR-WS*, 2025.

[18] A. Mandviwalla, K. Ohshiro, and B. Ji. Implementing grover's algorithm on the ibm quantum computers. In *2018 IEEE International Conference on Big Data (Big Data)*, pages 2531–2537, 2018.

[19] Y. Zhang et al. A quantum framework for protein binding-site structure prediction on utility-level quantum processors. *Advanced Science*, 12(5):2513641, 2025.

[20] Youngseok Kim et al. Evidence for the utility of quantum computing before fault tolerance. *Nature*, 618(7965):500–505, 2023.

[21] Yudong Cao et al. Quantum algorithms for supervised and unsupervised machine learning. *arXiv preprint arXiv:2007.03661*, 2020.

[22] Arpita Vats, Rahul Raja, Ashish Kattamuri, and Abhinav Bohra. Quantum natural language processing: A comprehensive survey of models, architectures, and evaluation methods. In *Proceedings of the QuantumNLP: Integrating Quantum Computing*

with Natural Language Processing, pages 53–64. Association for Computational Linguistics, 2025.

[23] Wei Chen, James Smith, and Rahul Patel. Optimizing ansatz design in quantum generative adversarial networks using large language models. *arXiv preprint arXiv:2503.12884*, 2025.

[24] Vincent Russo and Andrea Mari. Quantum error mitigation by layerwise richardson extrapolation. *Physical Review A*, 110(6):062420, 2024.

[25] Mark Johnson, Li Zhao, and Min-Jae Kim. Extrapolative quantum error mitigation in continuous-variable systems beyond the training horizon. *arXiv preprint arXiv:2603.08548*, 2026.

[26] Carlos Garcia, Anna Lee, and Hao Wang. Evaluating retrieval-augmented generation variants for natural language-based sql and api call generation. *arXiv preprint arXiv:2602.07086*, 2026.

[27] Stefan Muller, Giovanni Rossi, and Gupta. Radar signal classification with quantum machine learning: Ansatz depth impact on expressibility. *MDPI*, 15(2):370, 2025.